

"Write here, write now": Spelling difficulty disrupts parallel planning in sentence production

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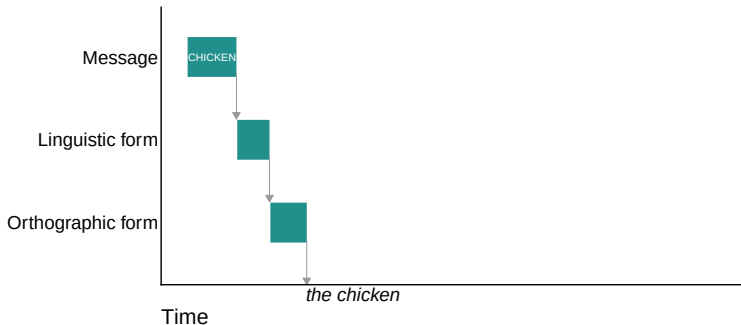
To what extent does difficulty with word spelling affect our ability to plan language in real time?

Planning language takes time!



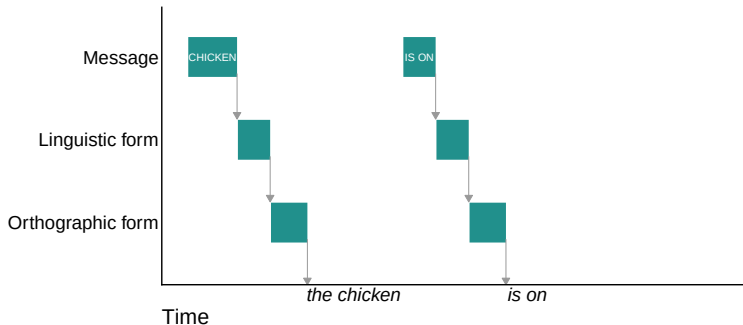
Serial view: Speaker / writers pause at linguistic edges to plan upcoming information.

Planning language takes time!



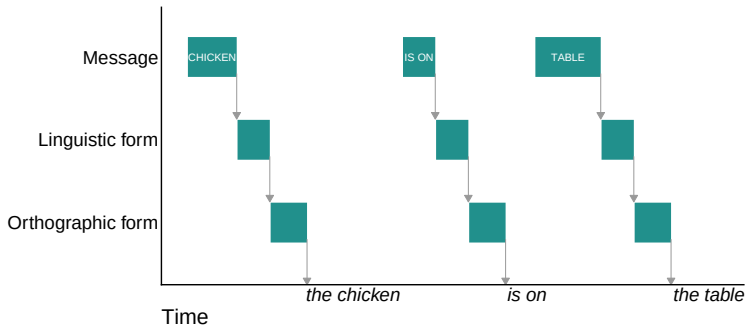
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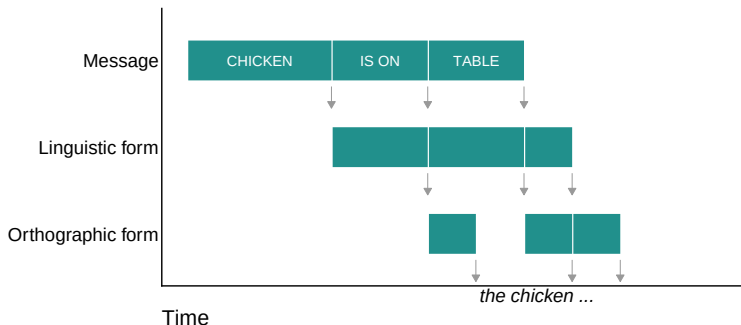
Planning language takes time!



Parallel view: Planning unfolds in parallel to output (Olive, [2014](#); Roeser et al., [2025](#); Van Galen, [1991](#)).

Pauses during production result from delays in information flow! Slow orthographic retrieval delays upstream processes.

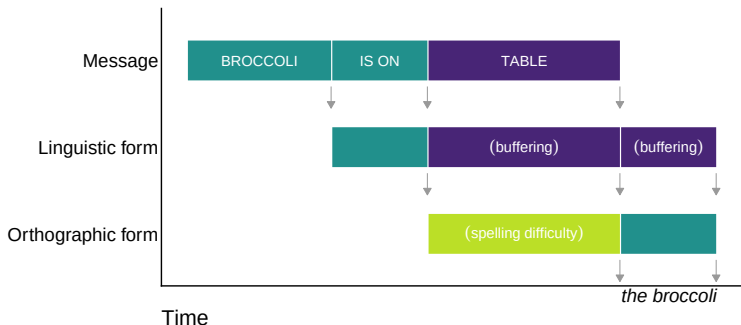
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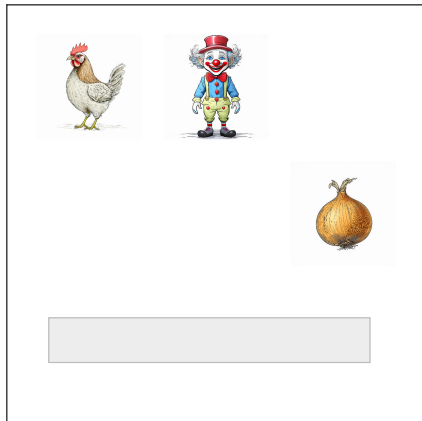
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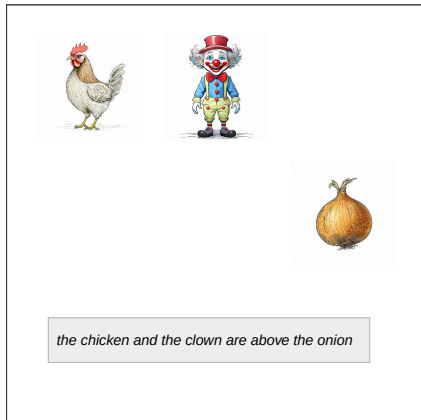
Pauses during production result from delays in information flow! Slow orthographic retrieval delays upstream processes.

Describe the arrangement from left to right!



Example screen (Roeser et al., [2019](#); Smith & Wheeldon, [1999](#))

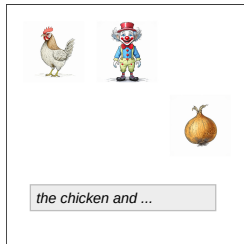
Describe the arrangement from left to right!



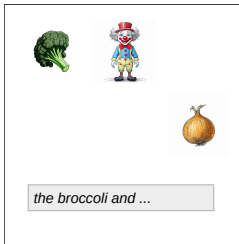
Example screen (Roeser et al., [2019](#); Smith & Wheeldon, [1999](#))

Design: Example item set

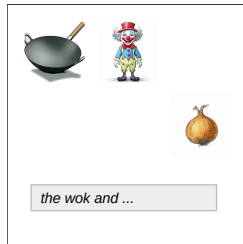
N1: easy spelling / long



N1 difficult spelling / long



N1: easy spelling / short



Prediction: Typing hesitations for N2 (*clown*) result from (1) when N1 spelling (*broccoli*) couldn't be retrieved in time and (2) when N1 length (*wok*) didn't allow for parallel planning.

- ▶ 36 items counterbalanced for prime structure (simple, conjoined subject) and N1 word type.
- ▶ 24 fillers with structures different from items.
- ▶ N1 word type manipulation: easy spelling (long), easy spelling (short), difficult spelling (long):
 - ▶ Long: ≥ 2 syllables
 - ▶ Difficult: spelling diversity $H_{spell} > 0.5$

What is “spelling diversity”?

Write the name of what you see!

What is “spelling diversity”?



What is “spelling diversity”?

Example summary: 75 / 97 participants used some spelling of “raccoon”.

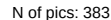
response	N_{spell}	Pr_{spell}	H_{spell}
raccoon	41	.55	1.31
racoön	30	.40	1.31
raccon	2	.03	1.31
raccoo	1	.01	1.31
racoo	1	.01	1.31

Note. H = diversity statistic; Pr = proportion of responses.

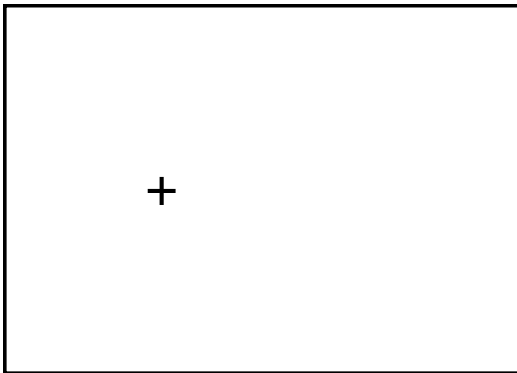
Frequency of a spelling used relative to the number different spellings:

$$H_{spell} = \sum_{i=1}^K p_i \times \log_2\left(\frac{1}{p_i}\right),$$

where k is the number of spellings for a picture name; p_i is the proportion of participants producing the i^{th} spelling (Lachman, 1973; Torrance et al., 2018).



Primed image description task



Primed image description task



Listen carefully:

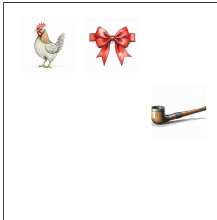
Primed image description task

"The chicken and the bow are above the pipe."

Listen carefully:

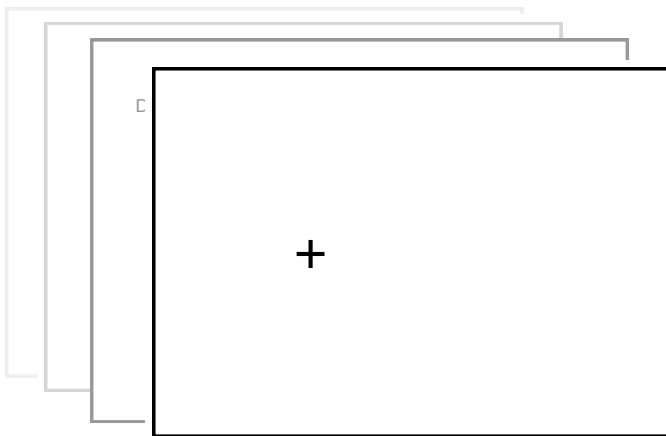
Primed image description task

Does the image on the screen match what you heard?

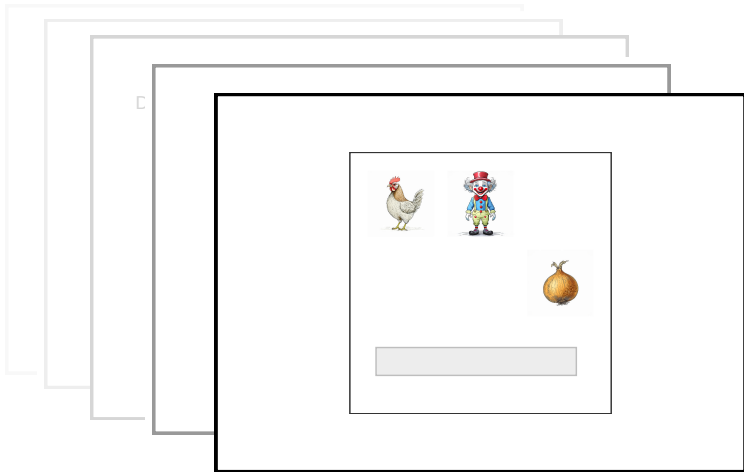


Press Y (Yes) or N (No)

Primed image description task



Primed image description task



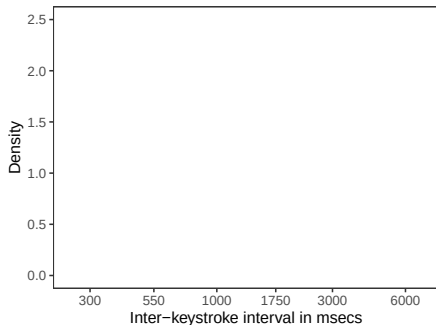
Primed image description task

C



the chicken and ...

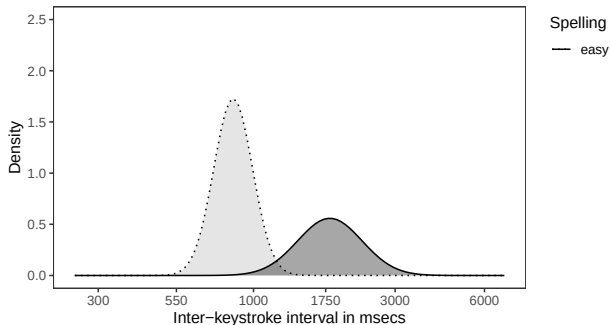
Data analysis: typing hesitations



Probability of typing hesitations were extracted from inter-keystroke intervals using Bayesian mixed-effects mixture model (Roeser et al., 2024, 2025).

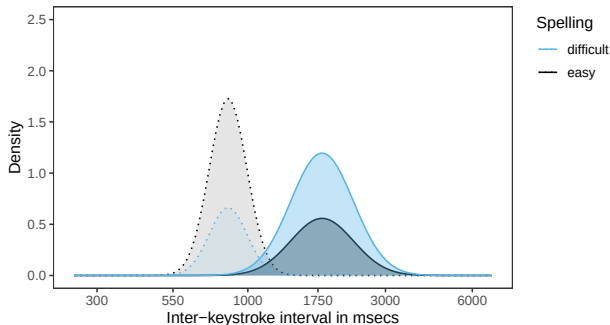
Tutorial: <https://rpubs.com/jensroes/mixture-models-tutorial>.

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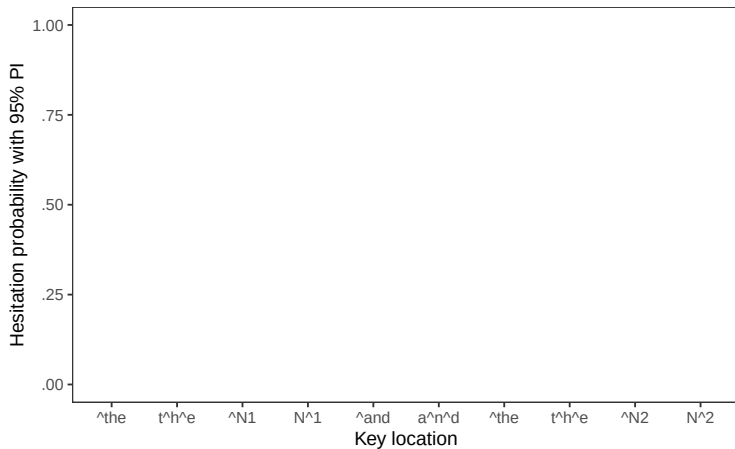
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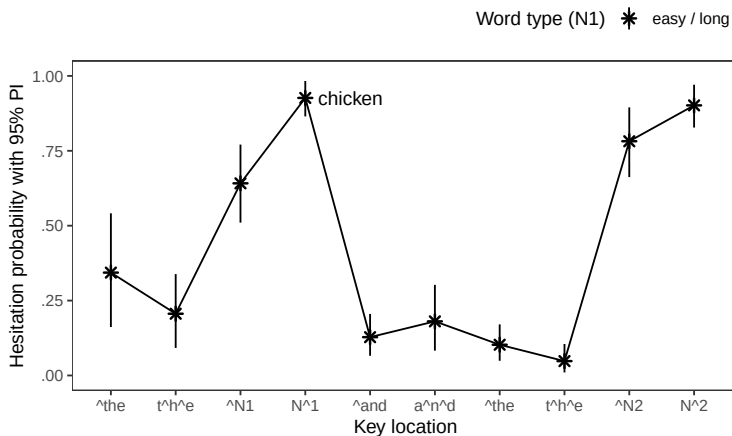
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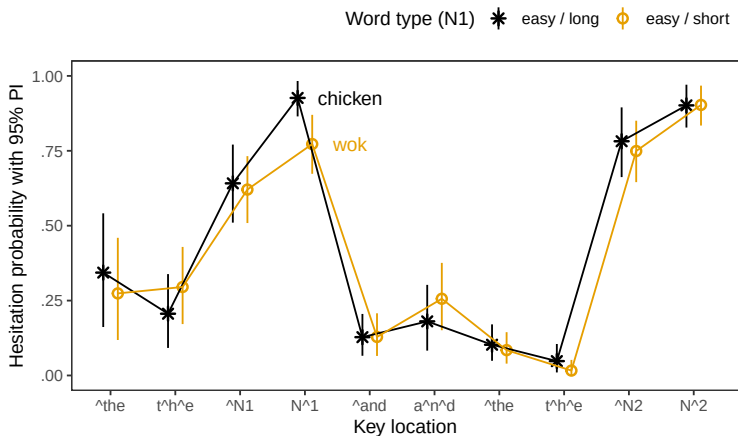
Experiment 1: $N = 56$ ppts; 6 cb lists



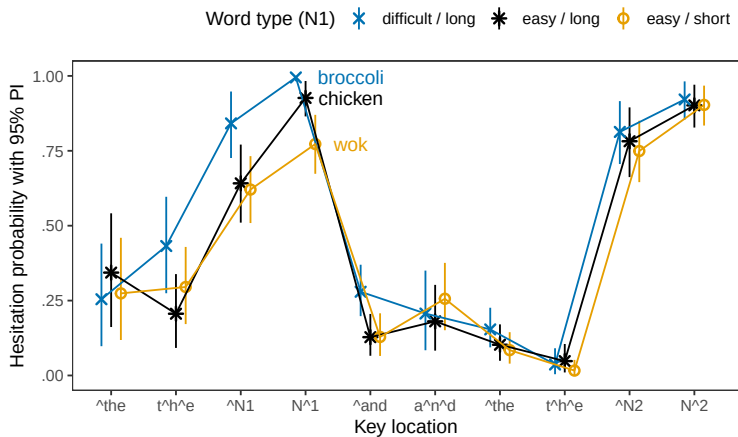
Experiment 1: $N = 56$ ppts; 6 cb lists



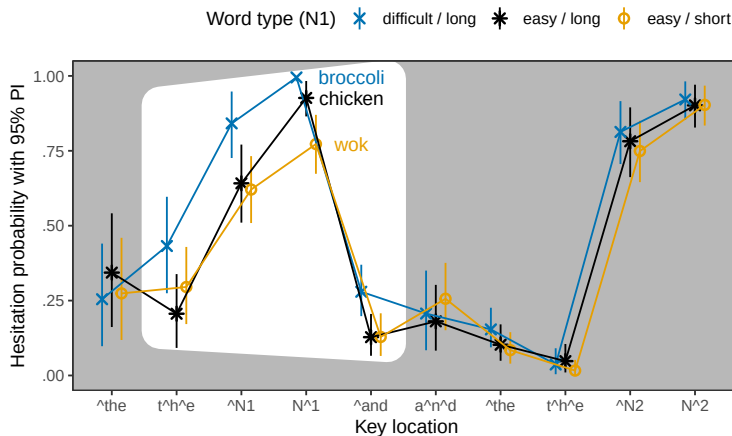
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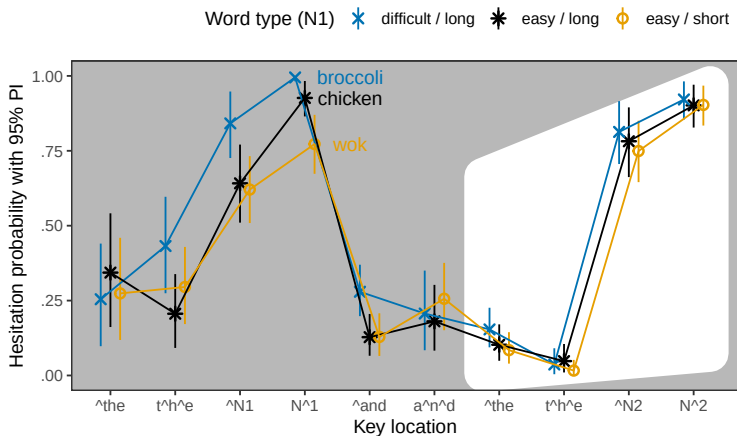
Experiment 1: $N = 56$ ppts; 6 cb lists



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Experiment 2

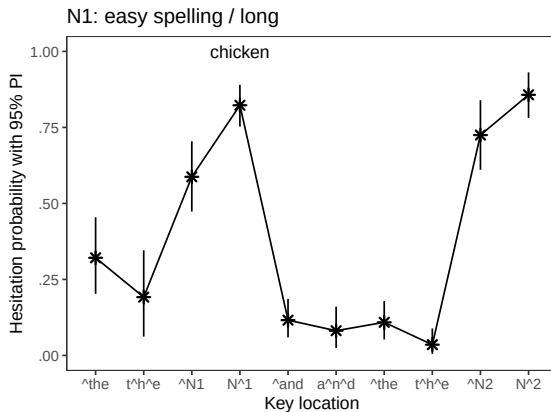
Does typing and the buy writers enough time to avoid hesitations before N2?

Additional prime condition; two target structures:

- ▶ With N2 determiner: *The N1 and the N2 are above the N3*
- ▶ Without N2 determiner: *The N1 and N2 are above the N3*

Prediction: If short distance between nouns constraints parallel planning, we expect more hesitations before N2 when *the* was omitted.

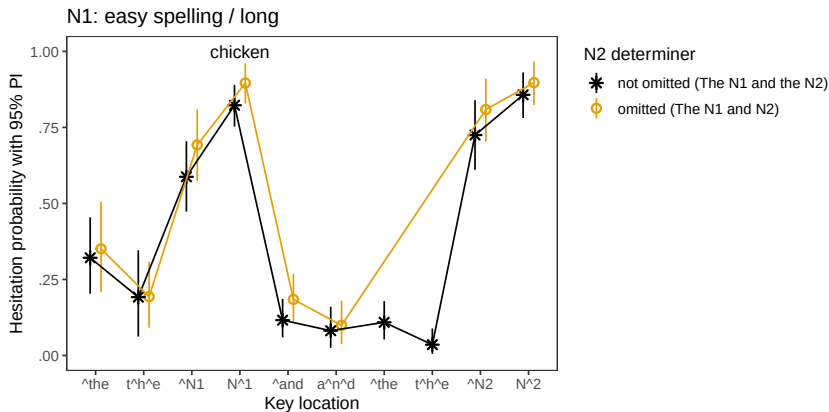
Experiment 2: $N = 80$ ppts; 9 cb lists



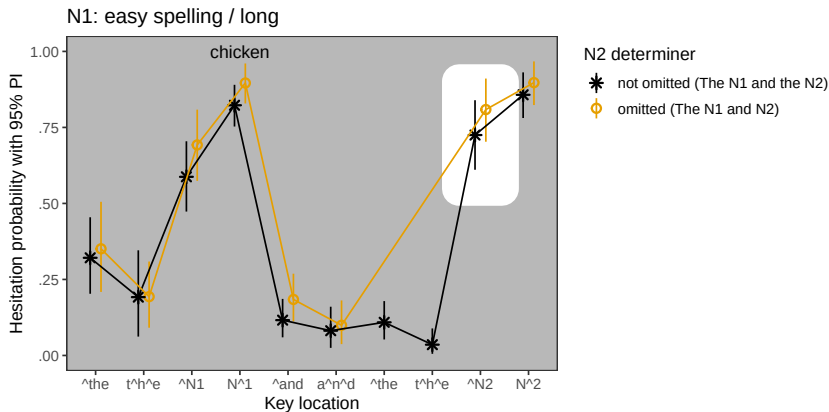
N2 determiner

* not omitted (The N1 and the N2)

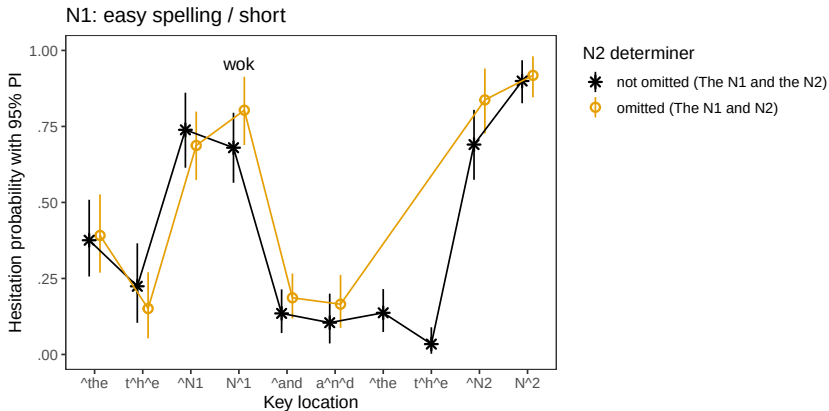
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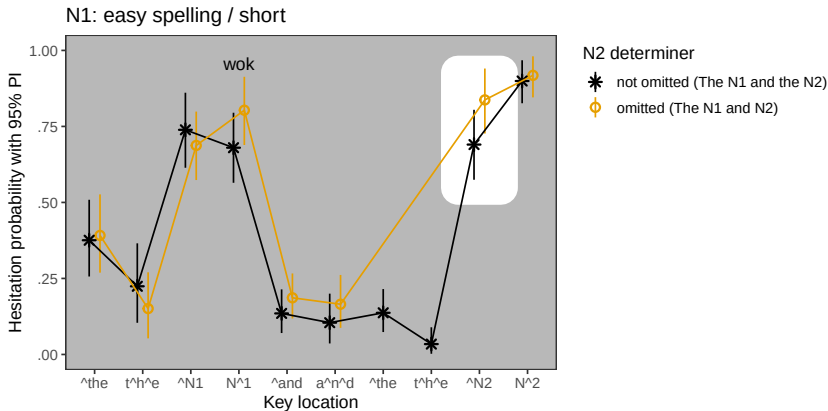
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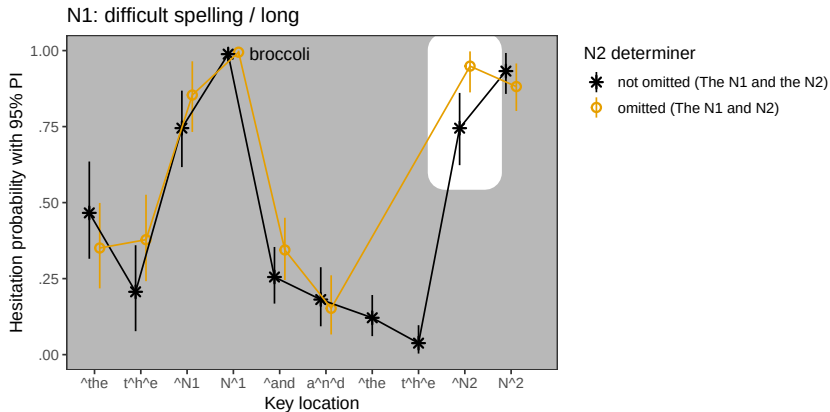


Experiment 2: $N = 80$ ppts; 9 cb lists





Experiment 2: $N = 80$ ppts; 9 cb lists



Experiment 3

Are hesitations related to delayed N2 retrieval?

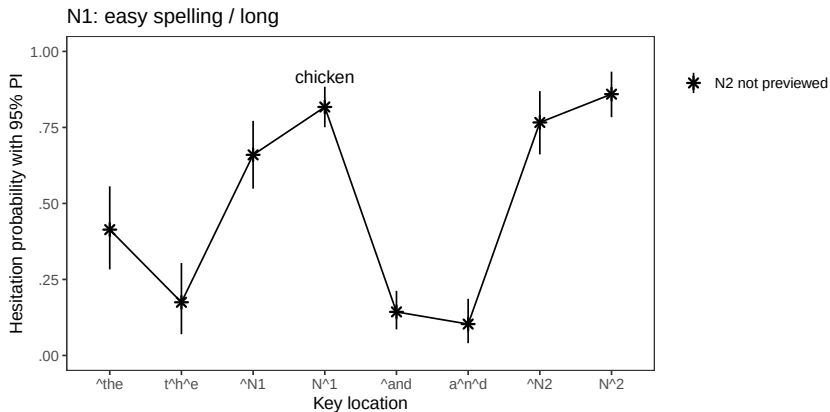
Prime with (auditory) N2 preview:

- ▶ Previewed: *The N1 and clown are above the pipe*
- ▶ Not previewed: *The N1 and bow are above the pipe*

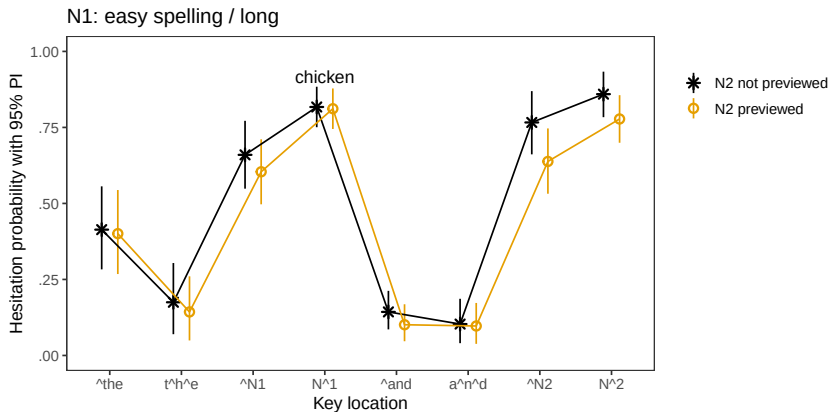
Target: *The N1 and clown are above the onion*

Prediction: If spelling difficulty delayed N2 retrieval, preview will reduce associated hesitations.

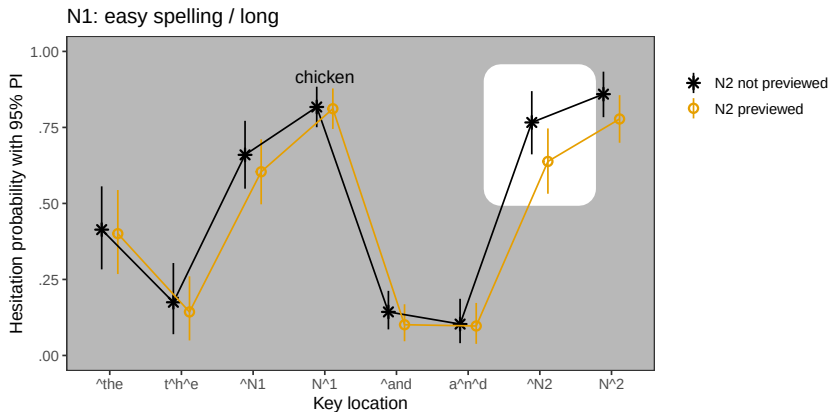
Experiment 3: $N = 96$ ppts; 12 cb lists



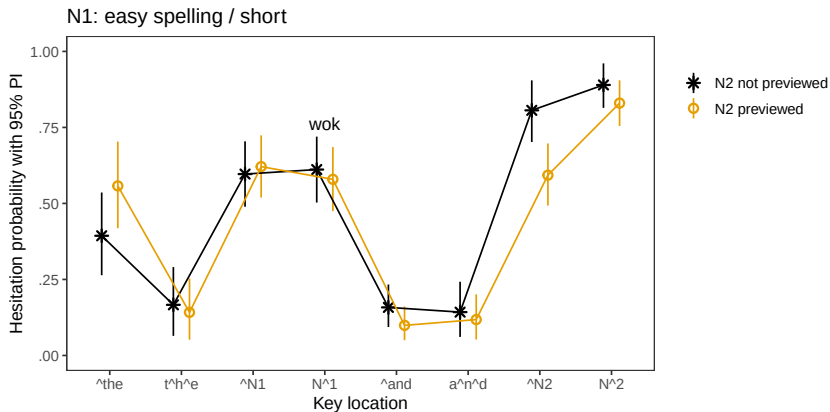
Experiment 3: $N = 96$ ppts; 12 cb lists



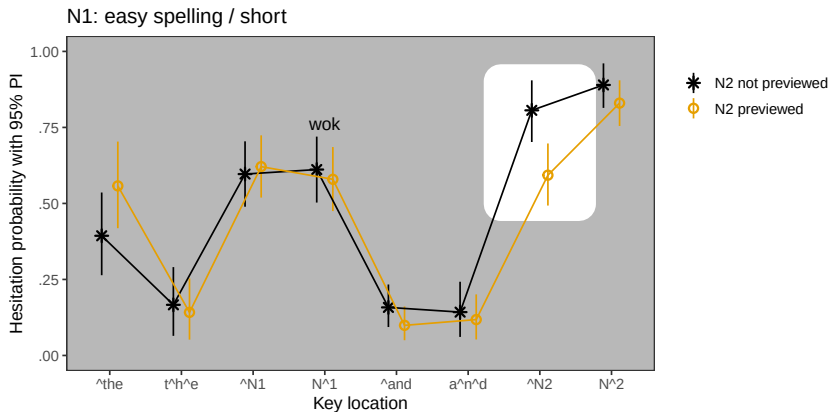
Experiment 3: $N = 96$ ppts; 12 cb lists



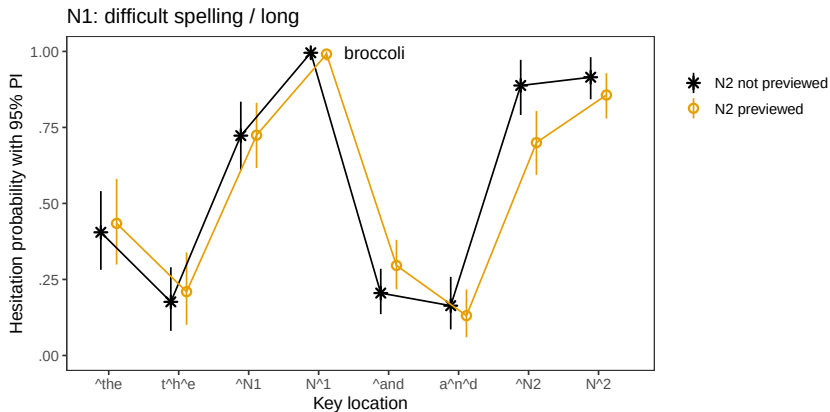
Experiment 3: $N = 96$ ppts; 12 cb lists



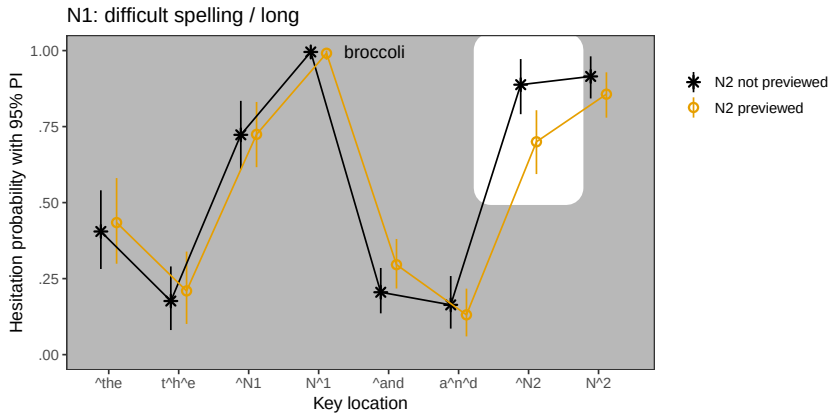
Experiment 3: $N = 96$ ppts; 12 cb lists



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Experiment 3: $N = 96$ ppts; 12 cb lists



- ▶ Hesitations related to N1 spelling affect N2 retrieval when there was not enough time between N1 and n2 for parallel lexical retrieval of N2.
- ▶ Long and / or functional words – e.g. *and the* – provide time for parallel lexical retrieval (see also Griffin, 2003).
- ▶ Written sentence planning unfolds in parallel to production: difficulty with spelling creates a planning bottleneck with knock-on effects on higher level planning.
- ▶ Serial models of language planning cannot explain these results (Flower & Hayes, 1980; Hayes, 2012).

Thanks for listening!

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Funded by UKRI ESRC (ES/W011832/1) – Project title: "Can you use it in a sentence?": Establishing how word-production difficulties shape text formation

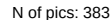
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References III

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Typing as mixture process

Inter-keystroke intervals were analysed in mixture models with shifted-lognormal distributions (Roeser et al., 2024, 2025).

$$(\log(iki_{ij}) - shift_i) \sim \theta_{condition} \times \mathcal{N}(\beta_{location} + \delta_{location} + u_i + w_j, \sigma_{e'}^2) + \\ (1 - \theta_{condition}) \times \mathcal{N}(\beta_{location} + u_i + w_j, \sigma_e^2)$$

where:

$$u_i \sim \mathcal{N}(0, \sigma_u^2)$$

$$w_j \sim \mathcal{N}(0, \sigma_w^2)$$

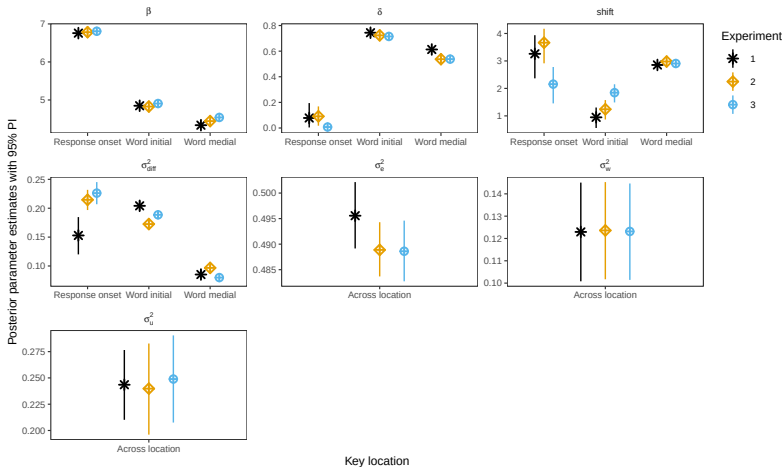
constraints:

$$\delta, \text{ (and all } \sigma\text{s)} > 0$$

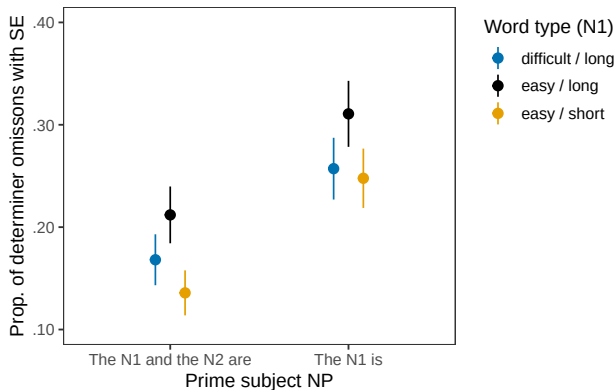
$$\sigma_{e'}^2 > \sigma_e^2$$

θ returns the proportion of hesitant inter-keystroke intervals for each key-location, prime and N1 condition.

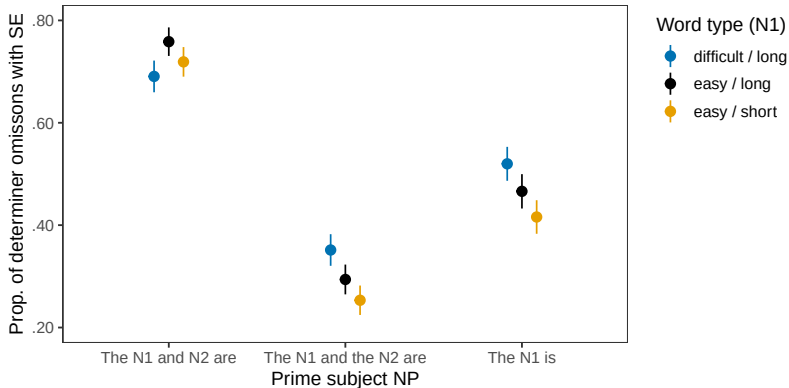
Most other mixture model parameter estimates



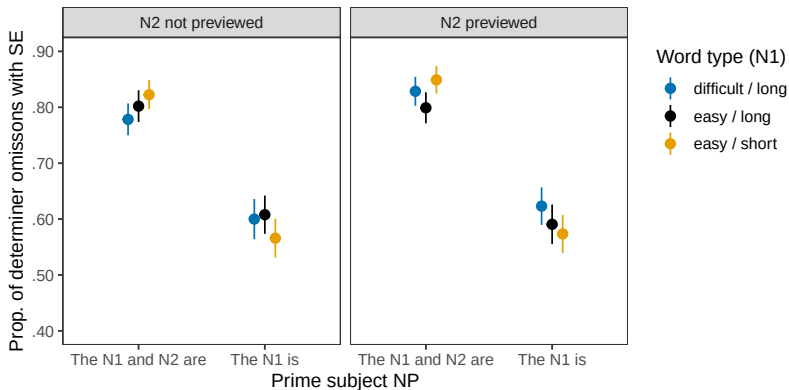
Experiment 1: determiner omissions before N2



Experiment 2: determiner omissions before N2



Experiment 3: determiner omissions before N2



Experiment 3: N2-initial hesitations

